

Biology 30 Final Exam Study Map & Quick Review Outline

Unit 1: What is Life? (Chapter 1)

1.1 Defining "Life"

- **The 7 Scientific Characteristics:** Know these core requirements (Refer to Textbook Section 1.1 for details):
 1. **Order:** Organized cellular hierarchy.
 2. **Reproduction:** Passing genetic material to offspring.
 3. **Growth & Development:** DNA-guided maturation.
 4. **Energy Processing:** Metabolism (converting food/sunlight to usable energy).
 5. **Response to Environment:** Reacting to dynamic stimuli.
 6. **Regulation / Homeostasis:** Keeping internal states stable.
 7. **Evolutionary Adaptation:** Population changes over generations.
- **Mnemonic Limitations:** Acronyms like **MRS GREN** have limits. Consider exceptions: fires "consume energy" and sterile mules "cannot reproduce," yet we classify only one as alive.

1.2 Cultural & Diverse Perspectives

- **Western Reductionism:** Breaking complex systems down into their smallest parts (e.g., studying organelles to understand the whole cell).
- **Indigenous Holism:** Emphasizing relationships and kinship (*wâhkôhtowin*) connecting all animate and inanimate things.
- **Animacy in Language:** In Cree (*nêhiyawêwin*), a rock (*asiniy*) is grammatically animate, recognizing its spiritual, ecological, and historical place in the environment.
- **Reciprocity:** Traditional protocols, such as offering tobacco when harvesting medicinal sweetgrass or sage, respect this web of life.

1.3 The History of Biogenesis

- **Abiogenesis (Spontaneous Generation):** The historical belief that life routinely arises from non-living matter (e.g., Aristotle's aphids from dew; Helmont's recipe for mice).
- **The Paradigm Shift:**
 - *Francesco Redi (1668):* Used open, sealed, and gauze-covered meat jars to show maggots come from flies, not decaying meat.
 - *Louis Pasteur (1859):* Used **Swan-Neck Flasks** to prove microbes exist in the air and do not generate spontaneously. The curved necks let air in but trapped dust/microbes until tilted.

1.4 Sub-Cellular "Gray Areas"

- **Viruses:** Infectious DNA or RNA in a protein capsid; require host machinery to replicate.

- **Viroids:** Infectious, naked, circular RNA loops that infect plants.
- **Prions:** Misfolded infectious proteins with no nucleic acids; cause healthy brain proteins to misfold.

Saskatchewan Connection: Chronic Wasting Disease (CWD) is a fatal prion disease in wild cervids (deer, elk, moose) in Saskatchewan. Prions bind to prairie soils and persist for years.

1.5 Cell Theory & Technology

- **The Three Pillars of Cell Theory:**
 1. All living things are made of one or more cells.
 2. The cell is the basic structural and functional unit of life.
 3. All cells come from pre-existing cells.

Saskatchewan Connection: The Canadian Light Source (CLS) synchrotron in Saskatoon produces ultra-bright light to help scientists analyze the molecular structures of agricultural plants, soils, and cell damage.

Unit 2: The Organization of Life (Chapters 2–4)

Chapter 2: Cell Structure & Energy Processes

2.1 Cell Types

Review the primary structural differences between prokaryotic and eukaryotic organisms:

Feature	Prokaryotes ("Before Nucleus")	Eukaryotes ("True Nucleus")
Nucleus	No (DNA is free-floating in the nucleoid region)	Yes (DNA is enclosed within a double-membrane envelope)
Size	Small (1–5 µm)	Larger (10–100 µm)
DNA Structure	Single, circular loop of DNA	Multiple, linear chromosomes
Organelles	Lack membrane-bound organelles	Specialized membrane-bound compartments
Examples	Bacteria and Archaea	Plants, Animals, Fungi, and Protists

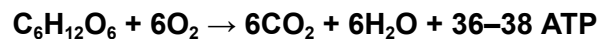
Saskatchewan Connection: *Rhizobium* bacteria (prokaryotes) form a symbiotic partnership with local pulse crops (lentils, peas) to fix atmospheric nitrogen into fertile soil nutrients.

2.2 Key Organelles

- **Endomembrane System:** Nucleus (DNA storage) → Rough ER (protein synthesis) → Smooth ER (lipid synthesis/detoxification) → Golgi Apparatus (sorting/packaging) → Vesicles (transport) → Lysosomes (digestion; animal cells only).
- **Energy & Boundaries:** Mitochondria (ATP generation via respiration), Chloroplasts (photosynthesis; plants only), Cell Wall (rigid plant cellulose, fungal chitin, or bacterial peptidoglycan), Cell Membrane (fluid phospholipid bilayer).

2.3 Biochemical Energetics

Part A: Cellular Respiration



1. **Glycolysis (Cytoplasm):** Splits glucose into pyruvate. Anaerobic. Yields 2 ATP and 2 NADH.
2. **Transition Reaction (Mitochondrial Matrix):** Converts pyruvate to Acetyl-CoA. Releases CO_2 .
3. **Citric Acid / Krebs Cycle (Mitochondrial Matrix):** Oxidizes Acetyl-CoA. Yields 2 ATP, 6 NADH, 2 FADH_2 , and releases CO_2 .
4. **Electron Transport Chain (Inner Membrane/Cristae):** High-energy electrons from carriers power proton (H^+) pumps, creating a gradient. Protons rush through **ATP Synthase** to generate ~32–34 ATP. Oxygen acts as the final electron acceptor, forming H_2O .
 - *Tracking Science:* Inhaled radioactive $^{18}\text{O}_2$ ends up in H_2O , not CO_2 .

Part B: Photosynthesis



1. **Light-Dependent Reactions (Thylakoids/Grana):** Solar energy excites chlorophyll. Water is split (**photolysis**), releasing O_2 gas and producing ATP and NADPH.
2. **Light-Independent Reactions / Calvin Cycle (Stroma):** Uses ATP and NADPH to fix gaseous CO_2 into G3P (sugars) using the enzyme **Rubisco**.

2.4 Macromolecules

- **Carbohydrates:** Monosaccharides (glucose) form polysaccharides (energy-storing starch/glycogen, structural cellulose).
- **Lipids:** Hydrophobic. Phospholipids are *amphipathic* (polar phosphate head, non-polar fatty acid tails), forming the cell membrane.

Saskatchewan Connection: Saskatchewan flax crops are rich in Omega-3 fatty acids, which keep cell membranes fluid and flexible.

- **Proteins:** Polypeptides made of 20 amino acids. Folding levels: **Primary** (sequence), **Secondary** (α -helices/ β -sheets), **Tertiary** (3D subunit shape), and **Quaternary** (multiple subunits).
- **Nucleic Acids:** DNA (double-stranded, deoxyribose, A-T/C-G base pairs) and RNA (single-stranded, ribose, uracil replaces thymine).

2.5 Membrane Transport

- **Passive (No Energy, Down Gradient):** Simple diffusion (small molecules like O_2), facilitated diffusion (needs transport proteins), and **osmosis** (diffusion of water).
 - *Osmotic Solutions:* **Hypertonic** (cell shrivels due to water exiting), **Hypotonic** (cell swells/lyses due to water entering), and **Isotonic** (equal exchange).

Saskatchewan Connection: Fertilizer Burn occurs when excess nitrogen fertilizer creates a hypertonic soil environment, drawing water out of crop roots via osmosis.

- **Active (Requires ATP, Up Gradient):** Protein pumps, endocytosis (phagocytosis/pinocytosis), and exocytosis.

Chapter 3: Biological Classification

3.1 Classification Systems

- **Functional:** Grouping by practical utility. (e.g., classifying Wild Oats as a "Noxious Weed" under the Saskatchewan Seeds Act).
- **Morphological:** Grouping by physical similarities (e.g., grouping birds and bats by wings).
- **Phylogenetic:** Grouping by shared evolutionary ancestry and genetic relationships.

3.2 Taxonomic Worldviews

- **Linnaean System:** Hierarchical, nested categories (pyramid design).
- **Medicine Wheel Framework:** Holistic, circular organization mapping life by season, direction, and elements (earth, air, fire, water).
- **Wâhkôhtowin:** Kinship-based understanding where humans are equal relations in the circle of life, not masters at the top of a pyramid.

3.3 Standardized Linnaean Taxonomy

- **Binomial Nomenclature:** Two-part Latin names: *Genus* (capitalized) and *specific epithet* (lowercase). Must be *italicized* (typed) or underlined (handwritten), e.g., *Bison bison bison*.
- **Taxonomic Ranks:** Domain → Kingdom → Phylum → Class → Order → Family → Genus → Species

- **Three Domains:** Bacteria (peptidoglycan walls), Archaea (extremophile prokaryotes), and Eukarya (all eukaryotes).
- **Dichotomous Key:** Identification tool using successive, opposing physical choices.
- **Cladogram:** Evolutionary tree showing relationships based on shared derived characters.

Chapter 4: Comparative Anatomy & Physiology

4.1 Homeostasis

- **Negative Feedback Loop:** Homeostatic pathway where a change triggers a response that opposes the initial stimulus (e.g., sweating to reduce body temperature).
- **Thermoregulation:** **Ectotherms** rely on behavior/sunlight for heat; **Endotherms** generate heat internally via metabolism.

Saskatchewan Connection: The Wood Frog survives freezing prairie winters by flooding its blood with glucose, acting as a natural antifreeze (cryoprotectant) to protect its cells.

- **Osmoregulation:** Balancing water/salt levels (e.g., kidneys filtering blood in animals; transpiration control in plants).

4.2 Cross-Kingdom Anatomical Solutions

- **Plants:** Stationary autotrophs; transport water via rigid **xylem** (transpirational pull) and sugars via living **phloem**. Gas exchange occurs through **stomata**.
- **Fungi:** Decomposers; use an underground network of thread-like **hyphae** (mycelium) to secrete digestive enzymes externally and absorb nutrients.
- **Animals:** Mobile heterotrophs; require internal organ systems (lungs/gills with high surface area, closed circulatory networks) to deliver nutrients and oxygen rapidly.

4.3 Structural Evidence

- **Homologous:** Shared evolutionary origin, different modern functions (e.g., human arm and bat wing).
- **Analogous:** Different evolutionary origins, same function due to similar environmental pressures (e.g., bird wing and insect wing).

4.4 Research Ethics & Models

- **Model Organisms:** Common species studied in labs to understand general biology (e.g., *E. coli*, fruit flies, mice).
- **The 3Rs of Research:** **Replacement** (using non-animal models), **Reduction** (using fewer animals), and **Refinement** (minimizing pain/distress).
- **Indigenous Research Principles:** Relational research that views animals as kin and teachers, requiring respect, permission (protocols like offering tobacco), and **reciprocity** to benefit the ecosystem.

4.5 Reproduction & Defense

- **Asexual vs. Sexual:** Asexual is fast and energy-efficient but produces clones vulnerable to single disasters. Sexual reproduction costs energy but yields genetic diversity crucial for adaptation.

Saskatchewan Connection: Trembling Aspen groves propagate asexually through root suckers, forming massive connected clonal colonies that can rapidly re-establish a canopy after a prairie fire.

- **r/K Selection:** **r-selected** species produce many offspring with no care (e.g., Northern Pike); **K-selected** species produce few offspring with high investment (e.g., Bears).

Saskatchewan Connection: Crop scientists breed wheat strains with specific resistance genes to protect crops from wheat rust fungus.

Unit 3: Biological and Molecular Inheritance (Chapters 5–6)

Chapter 5: Biological Inheritance

5.1 Classical Genetics & Terminology

- **Definitions:** Gene (heritable unit), Allele (gene variant), Genotype (genetic makeup, e.g., *BB*, *Bb*, *bb*), Phenotype (expressed trait, e.g., black coat vs. red coat in Angus cattle).
- **Monohybrid F₁ & F₂ Crosses:** Crossing purebred parents (*PP* × *pp*) yields 100% heterozygous F₁ offspring (*Pp*). Selfing the F₁ generation (*Pp* × *Pp*) yields a **1:2:1 genotypic ratio** (1 *PP* : 2 *Pp* : 1 *pp*) and a **3:1 phenotypic ratio** (3 Purple : 1 White).
- **Dihybrid Cross:** Crossing double heterozygotes (*SsPp* × *SsPp*) shows the **Law of Independent Assortment**, producing a classic **9:3:3:1 phenotypic ratio**.

5.2 Non-Mendelian Inheritance

- **Incomplete Dominance:** Heterozygote shows an intermediate, blended phenotype (e.g., pink flowers from red and white parents).
- **Codominance:** Both alleles are fully and independently expressed in the heterozygote (e.g., roan horses with both red and white hairs).
- **Multiple Alleles:** More than two alleles exist in a population for a single gene (e.g., ABO blood system).
- **Sex-Linked (X-Linked):** Genes carried on the X chromosome. Males (XY) are hemizygous and express recessive traits more frequently (e.g., hemophilia).
- **Polygenic Traits:** Continuous variation controlled by many genes working together (e.g., human skin color).

Saskatchewan Connection: Wheat kernel color is polygenic, controlled by three separate genes (A, B, and C). The darkness of the red pigment depends on the total number of dominant alleles present.

5.3 Human Blood Genetics

Analyze the inheritance patterns of the human ABO blood typing system:

Phenotype (Blood Type)	Genotype(s)	Surface Antigens Present	Transfusion Notes
Type A	$I^A I^A$ or $I^A i$	A antigen	Can receive Type A or O
Type B	$I^B I^B$ or $I^B i$	B antigen	Can receive Type B or O
Type AB	$I^A I^B$	Both A and B antigens	Universal Recipient (can receive any ABO type)
Type O	ii	None	Universal Donor (can donate red cells to anyone)

Rh Factor: Simple dominance. Dominant positive allele (R) vs. recessive negative (r).

Saskatchewan Connection: **STARS Air Ambulance** carries universal **Type O-Negative (*ii rr*)** blood because it lacks A, B, and Rh surface antigens, avoiding transfusion reactions in any recipient.

5.4 Meiosis & Chromosomes

- **Meiosis:** Two-stage cell division that reduces a diploid (2n) parent cell into four genetically diverse haploid (n) gametes.
- **Genetic Variation:** Driven by **Crossing-Over** (homologous chromosome segment swapping during Prophase I) and **Independent Assortment** (random chromosome alignment during Metaphase I).
- **Nondisjunction:** Chromatids fail to separate in Anaphase I or II, creating abnormal gametes.
 - *Karyotype:* Chromosome photo map used to diagnose chromosomal mutations and conditions like **Down Syndrome (Trisomy 21)**.

5.5 Population Genetics

- **Genetic Drift:** Random chance events altering allele frequencies in small populations.
 - *Founder Effect:* A small group splits off to start a new population with limited genetic diversity.
 - *Bottleneck Effect:* A sudden disaster crashes a population size, severely restricting the gene pool.

Saskatchewan Connection: Plains Bison suffered a major human-caused genetic bottleneck in the late 1800s. Modern populations have recovered in number but carry very low genetic diversity.

Chapter 6: Molecular Genetics

6.1 DNA Structure

- **Historical Timeline:** Chargaff (A=T, C=G); Rosalind Franklin (Photo 51, double helix X-ray); Watson & Crick (built the first accurate double-helix model using Franklin's data).
- **The Double Helix:** Antiparallel strands (5' → 3' and 3' → 5') with a covalent sugar-phosphate backbone and hydrogen-bonded nitrogenous bases on the interior.

6.2 The Central Dogma

[DNA REPLICATION] ---> Copying the DNA blueprint (S-Phase) | v [TRANSCRIPTION] ---> DNA template transcribed to mRNA (Nucleus) | v [TRANSLATION] ---> mRNA codons matched with tRNA anticodons to assemble amino acids at the Ribosome

1. **DNA Replication:** Semi-conservative process in Interphase (S-phase). **Helicase** unzips strands; **DNA Polymerase** synthesizes new strands 5' → 3' using complementary base-pairing.
2. **Transcription (Nucleus):** **RNA Polymerase** builds a single-stranded mRNA molecule from a DNA gene template. Uracil (U) replaces Thymine (T) in RNA.
3. **Translation (Cytoplasm/Ribosome):** Ribosomes read mRNA **codons** (3-letter sequences). **tRNA** molecules bring specific amino acids by matching their **anticodons** to the mRNA. Polypeptides grow until a STOP codon is reached.

6.3 Mutations

- **Point Mutations (Substitutions):**
 - *Silent:* No change to amino acid sequence.
 - *Missense:* Changes one amino acid, altering protein function.
 - *Nonsense:* Introduces a premature STOP codon, yielding a non-functional protein.
- **Frameshift Mutations (Insertions/Deletions):** Shift the entire reading frame down the line, scrambling all downstream codons and ruining the protein.

Saskatchewan Connection: Random mutations in Saskatchewan Kochia weeds altered the binding site of herbicides, creating herbicide-resistant superweeds across the prairies.

6.4 Epigenetics

- **Chemical Tags:** Chemical modifications (like **DNA Methylation**) turn genes ON or OFF without changing the actual DNA sequence.

- **Intergenerational Trauma:** Environmental stressors alter epigenetic tags in ancestors, which can be inherited by offspring.
- **Indigenous Connection:** Epigenetics aligns with the traditional view that ancestral experiences live on in descendants, explaining how starvation policies of the 1880s and residential school trauma left stress-susceptibility biological markers in modern generations.

6.5 Biotechnology Tools

- **PCR:** Amplifies tiny DNA samples into billions of copies.
- **Restriction Enzymes:** "Molecular scissors" that cut DNA at specific palindromic sequences.
- **Gel Electrophoresis:** Separates DNA fragments by size using electricity. Negatively charged DNA travels toward the positive pole. **Short fragments move fast and far; long fragments move slowly.**

Saskatchewan Connection: Conservation officers use gel electrophoresis DNA fingerprinting to match confiscated meat with poached wildlife.

Unit 4: Principles, Processes, and Patterns of Evolution (Chapter 7)

7.1 Key Concepts & Darwin's Work

- **Evolution Defined:** A change in the allele frequencies of a *population* (never an individual) over generations.
- **Darwin's Observations (HMS Beagle):**
 - *Global variation:* Ecologically similar animals live in similar habitats worldwide (rheas vs. ostriches).
 - *Local variation:* Finches have beaks adapted to their specific island's food source.
 - *Temporal variation:* Fossil species resemble living versions in the same area.
- **Darwinian Mechanics:** Overproduction of offspring → struggle for existence → variation of traits → differential survival and reproduction of those with helpful adaptations.

7.2 Selection Pressures & Modes

- **Artificial Selection:** Humans breed organisms for specific traits (e.g., dog breeds).

Saskatchewan Connection: Prairie researchers used artificial selection to breed toxic erucic acid out of rapeseed plants, creating **Canola** (Canadian Oil, Low Acid).

- **LUCA (Last Universal Common Ancestor):** The common ancestor of all life on Earth (3.5–3.8 billion years ago). Shared biochemistry (DNA, ATP, universal code) supports the holistic worldview of *wâhkôhtowin* ("All My Relations") as a scientific fact.

- **Modes of Selection:**

1. **Directional:** Shift toward one extreme phenotype (e.g., peppered moths turning dark).
2. **Stabilizing:** Favors the intermediate phenotype, selecting against extremes (e.g., human birth weight).
3. **Disruptive:** Favors both extremes over the average (e.g., small and large beak sizes on islands with no medium-sized seeds).

7.3 Historical Figures

- **Lamarck:** Proposed inheritance of acquired characteristics (stretched giraffe necks). Mechanism was wrong, but correctly argued that species adapt to environments.
- **Malthus:** Showed population growth outpaces resources, creating competition.
- **Wallace:** Independently discovered natural selection, prompting Darwin to publish his work.
- **Wegener:** Proposed Continental Drift, explaining why identical fossils are split across modern oceans.
- **Gould:** Proposed **Punctuated Equilibrium**—long stasis interrupted by rapid bursts of evolution.

7.4 Speciation & Evolutionary Patterns

- **Speciation:** Requires **Reproductive Isolation** to split a gene pool:
 - *Geographic:* Physical separation (rivers, mountains).
 - *Temporal:* Breeding at different times/seasons.
 - *Behavioral:* Different mating calls or rituals.
- **Evolutionary Paths:**
 - *Divergent (Adaptive Radiation):* One ancestor splits into many diverse species (e.g., Galapagos Finches).
 - *Convergent:* Unrelated species develop similar analogous traits due to shared environments (e.g., shark and dolphin shapes).
 - *Coevolution:* Two species evolve in step with each other (e.g., pollinator and flower).

7.5 Lines of Evidence Supporting Evolution

1. **The Fossil Record & Radiometric Dating:** Tracks extinct transitional forms. Radioactive decay isotopes (e.g., Carbon-14) calculate exact years.
Saskatchewan Connection: **Scotty the T-Rex**, excavated near Eastend, SK, is one of the world's most complete dinosaur skeletons, demonstrating pre-extinction dinosaur anatomy.
2. **Homologies:** Structural skeletal blueprint sharing (e.g., mammalian limbs).
3. **Vestigial Structures:** Leftover, evolutionary remnants with no modern function (e.g., whale pelvis, human appendix).
4. **Embryology:** Shared early developmental structures (e.g., temporary gill slits in vertebrate embryos).
5. **Molecular Biology:** Comparing DNA and amino acid sequences (e.g., Cytochrome C) to mathematically map evolutionary closeness.